

Trees

1. Write an algorithm for counting the number of left external nodes in a binary tree, using the Binary Tree ADT.

2. How do you delete a node with two children in a binary search tree?

3. perfect binary tree:
full / Proper binary tree:
complete binary tree:
balanced binary tree:

4. Rule for Binary search Tree:

5. Rule for AVL Tree:

6. Draw a binary tree T such that
- a) each internal node stores one character
 - b) preorder traversal yields EXAMFUN
 - c) inorder traversal yields MAFXUEN

7. Design and write an algorithm to traverse and output the post order traversal of a binary tree.

8. Create a binary search tree with the input QWERTYUIOP

9. Insert QWERTYUIOP into an initially empty AVL Tree

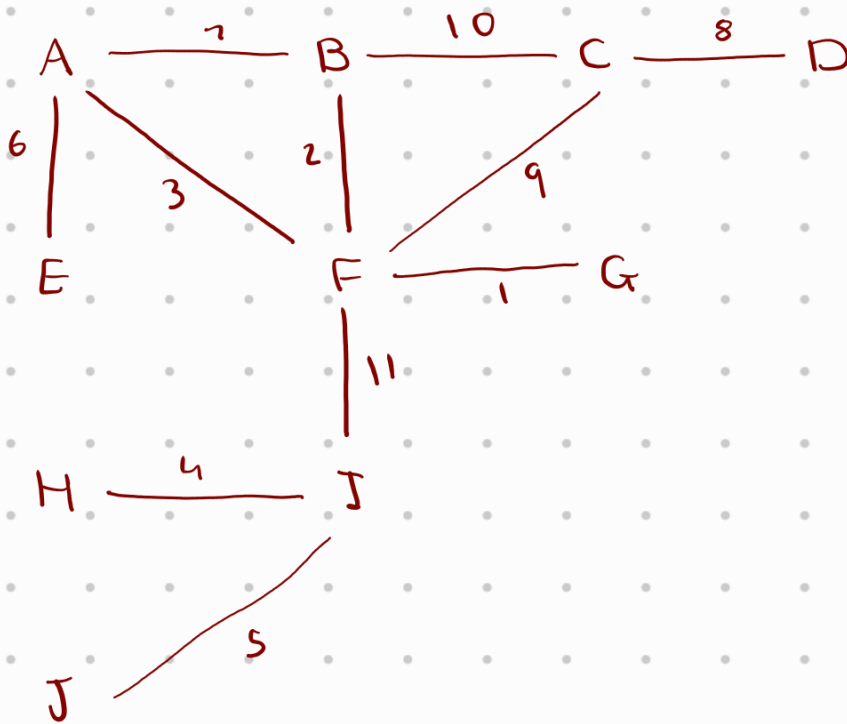
10. Represent the following array as a drawn tree, then convert it to a binary search tree, assuming you add elements to an initially empty BST by level order traversal of the given tree. What if you did a pre order traversal?

0	1	2	3	4	5	6
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E	X	N	A	U	M	F
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Graphs

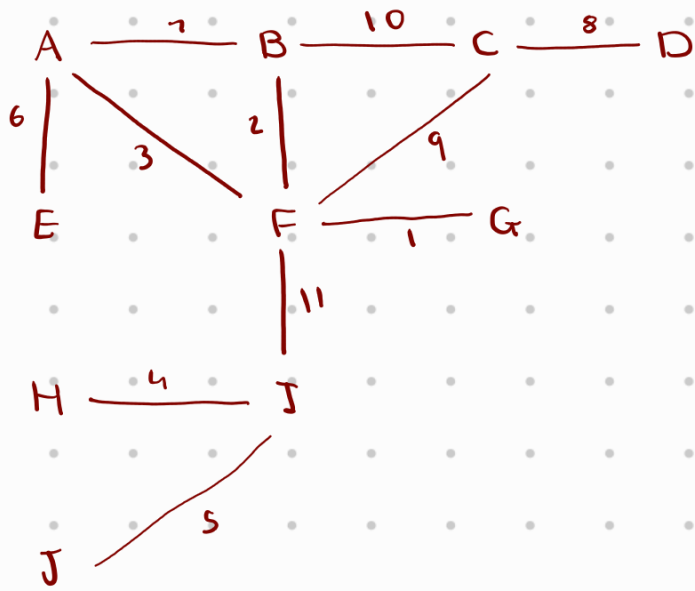
1. Graph Traversal (take smallest edge when applicable)



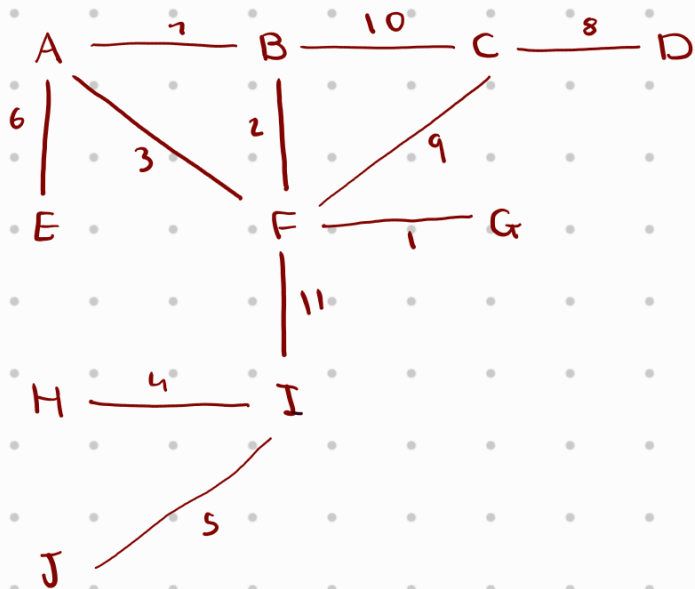
a) Starting at node A, output the breadth first traversal:

b) Starting at node B, output the depth first traversal:

2. Graph Algorithms



(a) Prim's Algorithm
starting at F, what
is the MST formed by
Prim's? (show edges in
the order they're
added to the MST)



(b) Kruskal's Algorithm

What is the MST formed by Kruskal's algorithm? Show edges in the order they're added.

3. Graph Terms

a) Incident:

b) Adjacent

c) Degree:

i) in-degree:

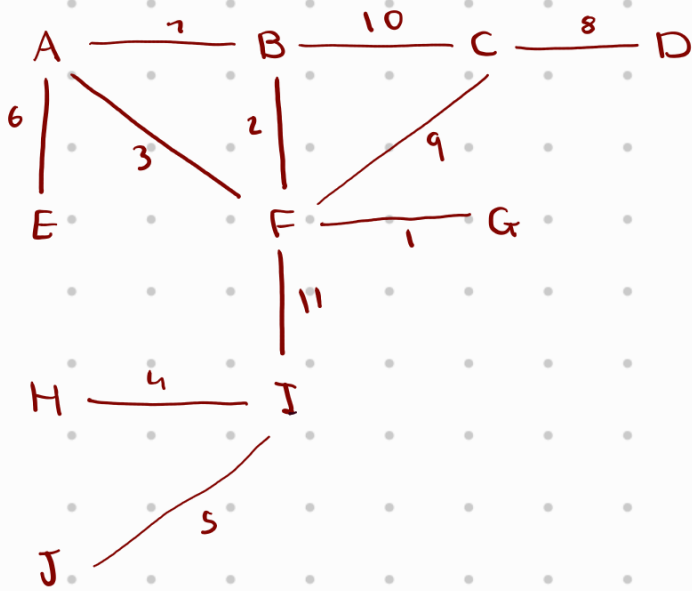
ii) out-degree:

d) self loop:

e) Parallel / multiple edges:

f) spanning tree vs. spanning subgraph:

4. Graph Representation



a) Represent the above graph as an adjacency list. Entries can be (adjacent vertex, edge weight).

Hashes

1. Add the values given, in order, to a hash table of capacity 11 using open addressing, using the hash function and collision handling method providing.

25 14 3 36 12 47

a) $h(x) = x \% 11$, linear probing

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

b) $h(x) = x \% 11$, quadratic probing

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

c) $h(x) = x \% 11$, double hashing

$$h_2(x) = 7 - (x \% 7)$$

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

2. a) Primary clustering:

b) Secondary clustering:

c) Load Factor:

3. Insert the following into an empty hash table using separate chaining:

25 14 3 36 12 47 58 8

using the hash function:

$$h(x) = x \% 11$$

Sorting note: stable means it preserves the relative order of equal/identical elements in the array.

1. Sort the following array in alphabetical order using merge sort. What are the time and space complexities? Is it stable? Show your steps.

["sit", "bit", "cat", "net", "dog", "map", "hop"]

2. Sort the following array in alphabetical order using quick sort. What are the time and space complexities? Is it stable? Show your steps, and use the last element as the pivot

["sit", "bit", "cat", "net", "dog", "map", "hop"]